

Model Test Coverage Report — COMP3211 PIM

Group: 89; member names and student IDs await human input.

Frozen product and test commit: 4d3508db0447a3ac6346a2a2789da89208f88fd5 (2026-09-30). **Evidence:** docs/reports/model-coverage-raw.txt, docs/reports/code-freeze.md, and docs/reports/integration-test-log.md. The course requires a separate model line-coverage report in the submitted source-code root; WP08 must place the final PDF there. This Markdown is the maintained report source.

1. Environment and method

WP04 ran on Windows 11 with Python 3.13.5 and 3.12.3. The test framework is Python's standard-library `unittest`. The coverage tool is `tools/model_coverage.py`, using the standard-library `trace` module. Python 3.11 was unavailable and remains unverified.

From the repository root, WP04 used:

```
$env:PYTHONPATH = "$PWD\src"
python -m unittest discover -s tests -t . -v
py -3.12 -m unittest discover -s tests -t . -q
python tools/model_coverage.py
python -m compileall -q src tests tools
```

The explicit `-t .` prevents `tests/model` from shadowing `src/model`. `tools/model_coverage.py` discovers and runs the full suite under `trace`, intersects traced line numbers with `trace._find_executable_linenos` for every `src/model/*.py` file, and prints covered/executable counts and percentages. No third-party coverage package or IDE coverage view generated the submitted figures.

2. Observed WP04 results

Check	Observed result
Python 3.13.5 full suite	33 tests; 0 failures; 0 errors.
Python 3.12.3 full suite	33 tests; 0 failures; 0 errors.
Python 3.13.5 model line coverage	411/430 executable lines; 95.58%.
Compile check	Exit code 0.
Manual CLI acceptance	Two separate processes created, searched, updated, deleted, saved, loaded, and continued with next ID 5; both exit codes 0.

The raw coverage output reports:

Model module	Covered/executable lines	Line coverage	Uncovered entries
model/__init__.py	0/1	0.00%	1
model/errors.py	4/5	80.00%	1
model/manager.py	92/97	94.85%	5
model/records.py	57/58	98.28%	1
model/search.py	224/233	96.14%	9
model/validation.py	34/36	94.44%	2
Total model	411/430	95.58%	19

The full raw text is retained in `docs/reports/model-coverage-raw.txt`; these values are copied exactly, without replacing the tool's denominator or rounding its module percentages. The reported coverage is **model-only**. The suite also contains controller, storage, and real-process acceptance tests, but those packages are outside the model numerator and denominator.

The acceptance tests launch a new CLI subprocess, whose executed lines are outside the parent trace session. Thus this report does not claim controller, storage, subprocess CLI, branch, condition, or mutation coverage.

3. Uncovered-line explanation

A read-only repeat of the frozen tool's line accounting showed the 19 unexecuted entries below. Python's `trace._find_executable_linenos` includes bookkeeping keys `0` or `None` in these files; they have no corresponding executable source statement. They account for nine denominator entries: one each in `model/__init__.py`, `errors.py`, and `records.py`, and two each in `manager.py`, `search.py`, and `validation.py`. The `model/__init__.py` file contains only a docstring, so its single counted entry is untraced. These entries are retained because WP04's raw calculation includes them.

The remaining ten entries correspond to actual frozen source lines:

Source	Untraced line(s)	Explanation
<code>model/manager.py::PIMManager.update</code>	70	The <code>field_name</code> non-string rejection branch was not reached by the 33 tests; CLI fields arrive as strings.
<code>model/manager.py::PIMManager.from_rows</code>	95	Direct non-array rows input to the model constructor was not exercised; file loading checks for an array in <code>storage/pim_file.py::load_pim</code> first.
<code>model/manager.py::PIMManager.from_rows</code>	101	Direct non-mapping record input was not exercised; the tests use malformed mappings and other file errors.
<code>model/search.py::OrCriterion._post_init__</code>	98	The tests construct a valid <code>OrCriterion</code> , but do not pass it a non-criterion operand.
<code>model/search.py::_tokenize</code>	185	The invalid-character branch was not reached by the sampled malformed search expressions.
<code>model/search.py::_Parser._accept</code>	205–209	This helper is present in the frozen parser but the iterative parse path never calls it; all five counted lines remain untraced.

The source-line numbers above are from freeze commit 4d3508d; line numbers in a later edited source could differ. The untraced branches show limits of the automated suite, not observed failures of those behaviors. The `_accept` helper is unused in the frozen product path. No official minimum coverage percentage is specified in the Project Description PDF.

4. Test scope and remaining verification

The 33-test suite includes `tests/model/test_records.py`, `test_manager.py`, `test_search.py`, and `test_search_integration.py`; it also includes controller, storage, and real-process acceptance tests. WP04's two-process manual session independently exercised the user-facing path, and its import audit found only standard-library or local source imports. `docs/reports/Requirements_Coverage.md` records requirement-level tests and direct-check gaps. Python 3.11 remains a requirement target without a passing run; neither the 3.12.3 nor the 3.13.5 result proves it.